Haoran Sun

B.S. Student in Mathematics, Fudan University
Research Intern, Computational Robotics Lab
Harvard University

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Education

Fudan University, Shanghai, China

B.S. in Mathematics and Applied Mathematics 9/2022 – 6/2027

Overall GPA: 3.77/4.00

GPA (Fall 2025–Present): 3.95/4.00

Related courses: Optimization (A); Numerical Solution of Differential Equations (A); Real Analysis (A); Advanced Algebra (A); Deep Learning (A); Mathematical Modeling (A); Ordinary Differential Equations (A)

University of California, Los Angeles, Los Angeles, CA

Exchange Student in Mathematics 9/2025 – 12/2025

GPA: 4.00/4.00

Graduate courses: MATH 273A Basic Optimization Theory (A+); MATH 275A Probability Theory (A+); MATH 269A Advanced Numerical Analysis (A)

Research Interests

I aim to develop *efficient computational algorithms* for large-scale optimization problems, particularly *GPU-accelerated methods*, together with *rigorous theoretical foundations*.

My current research focuses on: (a) developing scalable optimization algorithms that leverage insights from graph theory and numerical linear algebra while harnessing modern computing hardware, particularly GPUs; and (b) investigating the mathematical structure of optimization problems, particularly sparse moment relaxations, to understand when locally extracted atoms can be extended to global minimizers.

Experience

Harvard University, Cambridge, MA

Research Intern, Computational Robotics Lab 1/2026 – present

Advisor: Prof. Heng Yang

- Established theoretical guarantees for extracting globally extendable atomic candidates from sparse moment relaxations, even when only selected cliques satisfy simplicial regularity. Developed a progressive extraction method based on this theory and obtained optimal solutions on unicycle trajectory optimization. [1]
- Developed spectral deflation for factorization-free matrix filtering, with provable improvements in Newton–Schulz projection accuracy. Integrated spectral deflation into the Muon optimizer and ADMM for SDP, improving performance in language-model pretraining and large-scale SDP solving, where the improved solver converges to a KKT residual 50% lower than the original. [2]
- Accelerated the Python interface of SPOT, an open-source sparse polynomial optimization toolbox. Introduced NumPoly, a SymPy-free polynomial builder that speeds up polynomial conversion by 10× on average.

Lumin Capital, Shanghai, China

Quantitative Research Intern 7/2024 – 9/2024

- Built a text-driven statistical arbitrage strategy: extracted vector features from analyst research reports, trained XGBoost and GRU models to predict excess returns, and constructed factor signals achieving a 17% annualized return and a 1.6 Sharpe ratio.

Publications

[1] Haoran Sun. Global Extendability of Local Atoms in Sparse Moment Relaxations. arXiv:2610.17371, 2026. (arXiv)

[2] Haoran Sun, Shucheng Kang, and Heng Yang. Spectral Deflation for Factorization-Free Matrix Filtering in Muon and Semidefinite Programming. arXiv:2609.21102, 2026. (arXiv)

Software

SPOT, Sparse Polynomial Optimization Toolbox
An open-source, high-performance C++ toolbox with Python and MATLAB interfaces for solving polynomial optimization problems through sparse Moment-SOS relaxations. SPOT exploits correlative and term sparsity to substantially reduce SDP sizes and computational costs. It supports automatic and user-defined sparsity structures, flexible relaxation configurations, and solution extraction.

Teaching

Teaching Assistant, Calculus, Fudan University — Spring 2025
International School of Finance

Honors and Awards

William Lowell Putnam Mathematical Competition — 2026
Placed 113th (top 3%) of all participants; ranked 1st at UCLA (Basil Gordon Prize, $1,000)

Silver Medal, Chinese Mathematical Olympiad (CMO) — 2021

First Prize Scholarship (top 5%), Fudan University — 2025

First Prize (top 1% of teams), East China Mathematical Modelling Competition — 2024

Outstanding Student Leader, Fudan University — 2023

Skills

Programming: Python, C++, CUDA, MATLAB
Math: Semidefinite Programming, Numerical Linear Algebra, Large-Scale Optimization, GPU-Accelerated Methods

Activities

Calisthenics (member of the Fudan University Calisthenics Association), Rock climbing, Chinese bamboo flute